

3-Mark Questions & Answers — Data Structures Module 4: Trees

Q1. Explain the three Depth-First Traversals of a Binary Tree with examples.

1. **Preorder (Root -> Left -> Right):** Visit root first, then left and right subtrees.
 2. **Inorder (Left -> Root -> Right):** Yields sorted ascending order for Binary Search Trees.
 3. **Postorder (Left -> Right -> Root):** Used for bottom-up node deletion and expression evaluation.
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Q2. Explain the four AVL Tree rotations with diagrams.

1. **LL Rotation:** Single Right Rotation at unbalanced node **z** (Left-Left insertion).
 2. **RR Rotation:** Single Left Rotation at unbalanced node **z** (Right-Right insertion).
 3. **LR Rotation:** Double rotation (Left rotation on child **y**, then Right rotation on **z**).
 4. **RL Rotation:** Double rotation (Right rotation on child **y**, then Left rotation on **z**).
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Q3. Explain the steps of Huffman Encoding algorithm.

1. Count character frequencies and insert each into a Min-Heap.
 2. Repeatedly extract two nodes with lowest frequencies, combine them under a new parent node with sum frequency, and reinsert.
 3. Assign binary **0** to left branches and binary **1** to right branches to generate optimal prefix-free variable-length codes.
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Q4. State the 5 properties of Red-Black Trees.

1. Every node is either RED or BLACK.
 2. The root node is always BLACK.
 3. All **NIL** leaves are BLACK.
 4. If a node is RED, both its children must be BLACK (No consecutive REDs).
 5. Every path from a node to descendant **NIL** leaves has the same Black-Height.
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